

Section 2.3 - Question 1

Why is f not a function from $\mathbb{R}$ to $\mathbb{R}$ if

a

$$f(x) = 1/x?$$

The function is not defined for $x = 0$.

b

$$f(x) = \sqrt{x}?$$

The function is not defined for $x < 0$.

c

$$f(x) = \pm \sqrt{(x^2) + 1}?$$

Even though the function is defined for all x , it is not a function from $\mathbb{R}$ to $\mathbb{R}$ because it is not a single-valued function. For each x , there are two possible values for $f(x)$.

Section 2.3 - Question 9

a $\text{ceiling}(3/4) = 1$

b $\text{floor}(7/8) = 0$

c $\text{ceiling}(-3/4) = 0$

d $\text{floor}(-7/8) = -1$

e $\text{ceiling}(3) = 3$

f $\text{floor}(-1) = -1$

g $\text{floor}(1/2 + \text{ceiling}(3/2)) = \text{floor}(1/2 + 2) = \text{floor}(2.5) = 2$

h $\text{floor}(1/2 * \text{floor}(5/2)) = \text{floor}(1/2 * 2) = \text{floor}(1) = 1$

Section 2.3 - Question 10

Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.

a

$$f(a) = b, f(b) = a, f(c) = c, f(d) = d$$

The function is **one-to-one** because each element in the domain is mapped to a unique element in the codomain.

b

$$f(a) = b, f(b) = b, f(c) = d, f(d) = c$$

The function is **not one-to-one** because both a and b are mapped to b .

c

$$f(a) = d, f(b) = b, f(c) = c, f(d) = d$$

The function **not one-to-one** because both a and d are mapped to d .

Section 2.3 - Question 11

Which functions in Exercise 10 are onto?

a

$$f(a) = b, f(b) = a, f(c) = c, f(d) = d$$

The function is onto because each element in the codomain is mapped to by an element in the domain.

b

$$f(a) = b, f(b) = b, f(c) = d, f(d) = c$$

The function is not onto because a is not mapped to by any element in the domain.

c

$$f(a) = d, f(b) = b, f(c) = c, f(d) = d$$

The function is not onto because a is not mapped to by any element in the domain.

Section 2.3 - Question 21

Give an explicit formula for a function from the set of integers to the set of positive integers that is

a

one-to-one, but not onto.

The function $f(x) = 2x$ is one-to-one because $f(x) = f(y)$ implies $x = y$. However, the function is not onto because there is no integer x such that $f(x) = 1$.

b

onto, but not one-to-one.

The function $f(x) = |x|$ is onto because for every positive integer y , there is an integer x such that $f(x) = y$. However, the function is not one-to-one because $f(x) = f(-x)$ for all x .

c

one-to-one and onto.

The function $f(x) = x$ is one-to-one because $f(x) = f(y)$ implies $x = y$. The function is also onto because for every positive integer y , there is an integer x such that $f(x) = y$.

d

neither one-to-one nor onto.

The function $f(x) = 1$ is neither one-to-one nor onto because $f(1) = f(2)$ and there is no integer x such that $f(x) = 2$.

Section 2.3 - Question 23

Determine whether each of these functions is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

a

$$f(x) = 2x + 1$$

The function is a bijection because it is both one-to-one and onto.

b

$$f(x) = x^2 + 1$$

The function is not a bijection because it is not onto. For example, there is no real number x such that $f(x) = 0$.

c

$$f(x) = x^3$$

The function is a bijection because it is both one-to-one and onto.

d

$$f(x) = (x^2 + 1)/(x^2 + 2)$$

The function is not a bijection because it is not one-to-one. For example, $f(1) = f(-1) = 2/3$.

Section 2.3 - Question 25

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and let $f(x) > 0$ for all $x \in \mathbb{R}$. Show that $f(x)$ is strictly decreasing if and only if the function $g(x) = 1/f(x)$ is strictly increasing.

Suppose $f(x)$ is strictly decreasing. Then for all $x, y \in \mathbb{R}$, if $x < y$, then $f(x) > f(y)$. Then $1/f(x) < 1/f(y)$, so $g(x)$ is strictly increasing.

Section 2.3 - Question 26a

Prove that a strictly increasing function from $\mathbb{R}$ to itself is one-to-one.

- Prove f is one-to-one by way of contradiction
- Assume that f is strictly increasing from $\mathbb{R}$ to itself
 - Then, for all $x, y \in \mathbb{R}$, if $x < y$, then $f(x) < f(y)$
- Assume that f is not one-to-one.
 - Then, there exist $a, b \in \mathbb{R}$ such that $a \neq b$ and $f(a) = f(b)$.
- Without loss of generality, assume $a < b$.
 - Then $f(a) < f(b)$, which is a contradiction.
 - Therefore, f is indeed one-to-one if it is strictly increasing.

Section 2.3 - Question 36

If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

- Yes, it follows that g is one-to-one.
- Suppose f and $f \circ g$ are one-to-one.
- Then for all $x, y \in R$, if $f(x) = f(y)$, then $x = y$.
- Then for all $x, y \in R$, if $f(g(x)) = f(g(y))$, then $g(x) = g(y)$.
- Therefore, g is one-to-one.

Section 2.3 - Question 38

Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbb{R}$ to $\mathbb{R}$.

- $f \circ g(x) = f(g(x)) = f(x + 2) = (x + 2)^2 + 1 = x^2 + 4x + 4 + 1 = x^2 + 4x + 5$
- $g \circ f(x) = g(f(x)) = g(x^2 + 1) = x^2 + 1 + 2 = x^2 + 3$

Section 2.3 - Question 45

Let f be a function from the set A to the set B . Let S be a subset of B . We define the inverse image of S to be the subset of A whose elements are precisely all preimages of all elements of S . We denote the inverse image of S by $f^{-1}(S)$, so $f^{-1}(S) = \{a \in A \mid f(a) \in S\}$.

Let $g(x) = \text{floor}(x)$. Find

a

$$g^{-1}(\{0\}).$$

$$g^{-1}(\{0\}) = \{x \mid \text{floor}(x) = 0\} = \{x \mid 0 \leq x < 1\}.$$

b

$$g^{-1}(\{-1, 0, 1\}).$$

$$\begin{aligned} g^{-1}(\{-1, 0, 1\}) &= \{x \mid \text{floor}(x) = -1 \text{ or } \text{floor}(x) = 0 \text{ or } \text{floor}(x) = 1\} \\ &= \{x \mid -1 \leq x < 0 \text{ or } 0 \leq x < 1 \text{ or } 1 \leq x < 2\} = \{x \mid -1 \leq x < 2\}. \end{aligned}$$

c

$$g^{-1}(\{x \mid 0 < x < 1\}).$$

$$g^{-1}(\{x \mid 0 < x < 1\}) = \{x \mid 0 < \text{floor}(x) < 1\} = \{x \mid 0 < x < 1\}.$$

Section 2.3 - Question 53a

Show that if x is a real number and n is an integer, then $x < n$ if and only if $\text{floor}(x) < n$.

Suppose $x < n$. Then $\text{floor}(x) \leq x < n$, so $\text{floor}(x) < n$.

Section 2.3 - Question 55

Prove that if n is an integer, then $n/2 = \lfloor n/2 \rfloor$ if n is even and $(n - 1)/2$ if n is odd.

Suppose n is even. Then $n = 2k$ for some integer k , so $n/2 = k = \lfloor n/2 \rfloor$.

Section 2.3 - Question 65

Draw the graph of the function $f(x) = |2x|$ from $\mathbb{R}$ to $\mathbb{R}$.

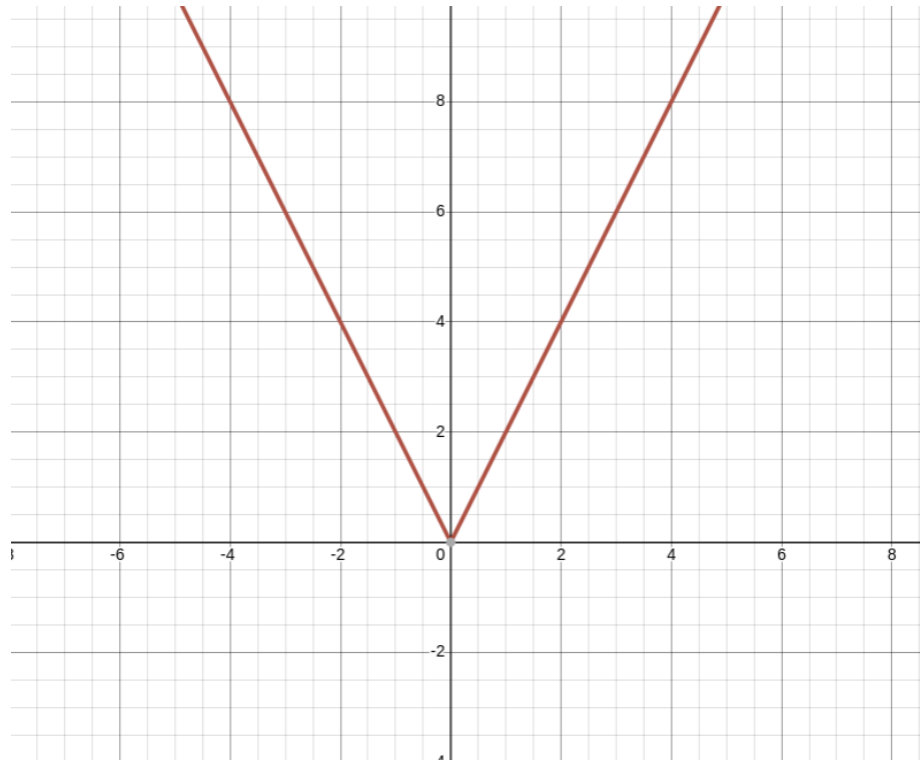


Figure 1: alt text

Section 2.3 - Question 71

Find the inverse function of $f(x) = x^3 + 1$.

$$f^{-1}(x) = (x - 1)^{1/3}$$

Section 2.3 - Question 78

Let x be a real number. Show that $\text{floor}(3x) = \text{floor}(x) + \text{floor}(x + 1/3) + \text{floor}(x + 2/3)$.

- Let $x = n + e$, where n is an integer and $0 \leq e < 1$.
- Case 1: $e < 1/3$
 - then $\text{floor}(x) = n$, $\text{floor}(x + 1/3) = n$, and $\text{floor}(x + 2/3) = n$,
so $\text{floor}(3x) = \text{floor}(x) + \text{floor}(x + 1/3) + \text{floor}(x + 2/3) = 3n$.
- Case 2: $1/3 \leq e < 2/3$
 - then $\text{floor}(x) = n$, $\text{floor}(x + 1/3) = n$, and $\text{floor}(x + 2/3) = n + 1$,
so $\text{floor}(3x) = \text{floor}(x) + \text{floor}(x + 1/3) + \text{floor}(x + 2/3) = 3n + 1$.
- Case 3: $2/3 \leq e < 1$
 - then $\text{floor}(x) = n$, $\text{floor}(x + 1/3) = n + 1$, and $\text{floor}(x + 2/3) = n + 1$,
so $\text{floor}(3x) = \text{floor}(x) + \text{floor}(x + 1/3) + \text{floor}(x + 2/3) = 3n + 2$.

Section 2.3 - Question 79

For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.

a

$$f : \mathbb{Z} \rightarrow \mathbb{R}, f(n) = 1/n$$

The domain is $\mathbb{Z}$, the codomain is $\mathbb{R}$, the domain of definition is $\mathbb{Z} - \{0\}$, and the set of values for which it is undefined is $\{0\}$. The function is not a total function.

b

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n/2$$

The domain is $\mathbb{Z}$, the codomain is $\mathbb{Z}$, the domain of definition is $\mathbb{Z}$, and the set of values for which it is undefined is the set of all odd integers. The function is not a total function.

c

$$f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}, f(m, n) = m/n$$

The domain is $\mathbb{Z} \times \mathbb{Z}$, the codomain is $\mathbb{Q}$, the domain of definition is $\mathbb{Z} \times (\mathbb{Z} - \{0\})$, and the set of values for which it is undefined is $\mathbb{Z} \times \{0\}$. The function is not a total function.

d

$$f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(m, n) = mn$$

The domain is $\mathbb{Z} \times \mathbb{Z}$, the codomain is $\mathbb{Z}$, the domain of definition is $\mathbb{Z} \times \mathbb{Z}$, and the set of values for which it is undefined is the empty set. The function is a total function.

e

$$f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(m, n) = m - n \text{ if } m > n$$

The domain is $\mathbb{Z} \times \mathbb{Z}$, the codomain is $\mathbb{Z}$, the domain of definition is $\mathbb{Z} \times \mathbb{Z}$, and the set of values for which it is undefined is the empty set. The function is a total function.

Section 2.4 - Question 1

Find these terms of the sequence $\{a_n\}$, where $a_n = 2 \cdot (-3)^n + 5^n$.

a

$$a_0$$

$$a_0 = 2 \cdot (-3)^0 + 5^0 = 2 \cdot 1 + 1 = 3$$

b

$$a_1$$

$$a_1 = 2 \cdot (-3)^1 + 5^1 = 2 \cdot (-3) + 5 = -6 + 5 = -1$$

c

$$a_4$$

$$a_4 = 2 \cdot (-3)^4 + 5^4 = 2 \cdot 81 + 625 = 162 + 625 = 787$$

d

$$a_5$$

$$a_5 = 2 \cdot (-3)^5 + 5^5 = 2 \cdot (-243) + 3125 = -486 + 3125 = 2639$$

Section 2.4 - Question 4

What are the terms a_0 , a_1 , a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals

a

- $a_0 = (-2)^0 = 1$
- $a_1 = (-2)^1 = -2$
- $a_2 = (-2)^2 = 4$
- $a_3 = (-2)^3 = -8$

b

- $a_0 = 3$
- $a_1 = 3$
- $a_2 = 3$
- $a_3 = 3$

c

- $a_0 = 7 + 4^0 = 8$
- $a_1 = 7 + 4^1 = 11$
- $a_2 = 7 + 4^2 = 23$
- $a_3 = 7 + 4^3 = 39$

d

- $a_0 = 2^0 + (-2)^0 = 1 + 1 = 2$
- $a_1 = 2^1 + (-2)^1 = 2 - 2 = 0$
- $a_2 = 2^2 + (-2)^2 = 4 + 4 = 8$
- $a_3 = 2^3 + (-2)^3 = 8 - 8 = 0$

Section 2.4 - Question 6b

List the first 10 terms of the sequence whose n th term is the sum of the first n positive integers

1, 3, 6, 10, 15, 21, 28, 36, 45, 55

Section 2.4 - Question 9e

Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions: $a_n = a_{n-1} + a_{n-3}$, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$

$$a_3 = a_2 + a_0 = 0 + 1 = 1$$

Section 2.4 - Question 13abdg

Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if

a

$$a_n = 0?$$

$$a_1 = 8a_0 - 16a_{-1} = 8 * 0 - 16 * 0 = 0$$

b

$$a_n = 1?$$

$$a_1 = 8a_0 - 16a_{-1} = 8 * 1 - 16 * 0 = 8$$

d

$$a_n = 4^n?$$

$$a_1 = 8a_0 - 16a_{-1} = 8 * 4 - 16 * 2 = 32$$

g

$$a_n = (-4)^n?$$

$$a_1 = 8a_0 - 16a_{-1} = 8 * (-4) - 16 * 0 = -32$$

h

$$a_n = n^2 4^n?$$

$$a_1 = 8a_0 - 16a_{-1} = 8 * 4 - 16 * 0 = 32$$

Section 2.4 - Question 15d

Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if $a_n = 7 \cdot 2^n - n + 2$.

$$a_1 = a_0 + 2a_{-1} + 2 - 9 = 7 \cdot 2 - 0 + 2 - 9 = 14 - 7 = 7$$

Section 2.4 - Question 19

Suppose that the number of bacteria in a colony triples every hour.

a

Set up a recurrence relation for the number of bacteria after n hours have elapsed.

$$a_n = 3a_{(n-1)}$$

b

If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?

$$a_{10} = 3^{10} * 100 = 5904900$$

Section 2.4 - Question 22

An employee joined a company in 2017 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.

a

Set up a recurrence relation for the salary of this employee n years after 2017.

$$a_n = a_{(n-1)} + 1000 + 0.05a_{(n-1)}$$

b

What will the salary of this employee be in 2025?

- $a_1 = 50000 + 1000 + 0.05 * 50000 = 51000 + 2500 = 53500$
- $a_2 = 53500 + 1000 + 0.05 * 53500 = 54500 + 2675 = 57175$
- $a_3 = 57175 + 1000 + 0.05 * 57175 = 58175 + 2858.75 = 61033.75$
- $a_4 = 61033.75 + 1000 + 0.05 * 61033.75 = 62033.75 + 3051.6875 = 65085.4375$
- $a_5 = 65085.4375 + 1000 + 0.05 * 65085.4375 = 66085.4375 + 3254.271875 = 69339.709375$
- $a_6 = 69339.709375 + 1000 + 0.05 * 69339.709375 = 70339.709375 + 3466.98546875 = 73806.69484375$
- $a_7 = 73806.69484375 + 1000 + 0.05 * 73806.69484375 = 74806.69484375 + 3690.3347421875 = 78497.0295859375$
- $a_8 = 78497.0295859375 + 1000 + 0.05 * 78497.0295859375 = 79497.0295859375 + 3949.851479296875 = 83446.881065234375$

c

Find an explicit formula for the salary of this employee n years after 2017.

$$\begin{aligned} a_n &= 50000 + n * 1000 + 0.05 * (50000 + (n - 1) * 1000) = 50000 + \\ &n * 1000 + 0.05 * 50000 + 0.05 * (n - 1) * 1000 = 50000 + n * 1000 \\ &+ 2500 + 0.05 * (n - 1) * 1000 = 52500 + n * 1000 + 0.05 * (n - 1) * 1000 \end{aligned}$$

explicit formula: $a_n = 52500 + n * 1000 + 0.05 * (n - 1) * 1000$

Section 2.4 - Question 29c

What is the value of these sums: `for (i = 1; i <= 10; i++)`
`{ 3 }`

$$3 * 10 = 30$$

Section 2.4 - Question 30a

What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

$\text{for } (i \text{ in } S) \{ i \}$

$$1 + 3 + 5 + 7 = 16$$

Section 2.4 - Question 31c

What is the value of each of these sums of terms of a geometric progression? for ($i = 2$; $i \leq 8$; $i++$) { $2 \cdot (-3)^i$ }

$$\begin{aligned} &2 \cdot (-3)^2 + 2 \cdot (-3)^3 + 2 \cdot (-3)^4 + 2 \cdot (-3)^5 + 2 \cdot (-3)^6 + 2 \cdot (-3)^7 + 2 \cdot (-3)^8 \\ &= 2 \cdot 9 - 2 \cdot 27 + 2 \cdot 81 - 2 \cdot 243 + 2 \cdot 729 - 2 \cdot 2187 + 2 \cdot 6561 \\ &= 18 - 54 + 162 - 486 + 1458 - 4374 + 13122 \\ &= 18630 - 18630 = 0 \end{aligned}$$

Section 2.4 - Question 32

Find the value of each of these sums

a

for ($i = 0; i \leq 8; i++$) { $1 + (-1)^i$ }

10

for ($i = 0; i < 8; i++$) { $1 + (-1)^i$ }

8

b

for ($i = 0; i \leq 8; i++$) { $3^i - 2^i$ }

$0 + 1 + 5 + 19 + 65 + 211 + 665 + 2059 + 6305 = 9330$

for ($i = 0; i < 8; i++$) { $3^i - 2^i$ }

3025

c

for ($i = 0; i \leq 8; i++$) { $23^i + 32^i$ }

7325

for ($i = 0; i < 8; i++$) { $23^i + 32^i$ }

21215

d

for ($i = 0; i \leq 8; i++$) { $2^{(i+1)} - 2^i$ }

511

for ($i = 0; i < 8; i++$) { $2^{(i+1)} - 2^i$ }

255